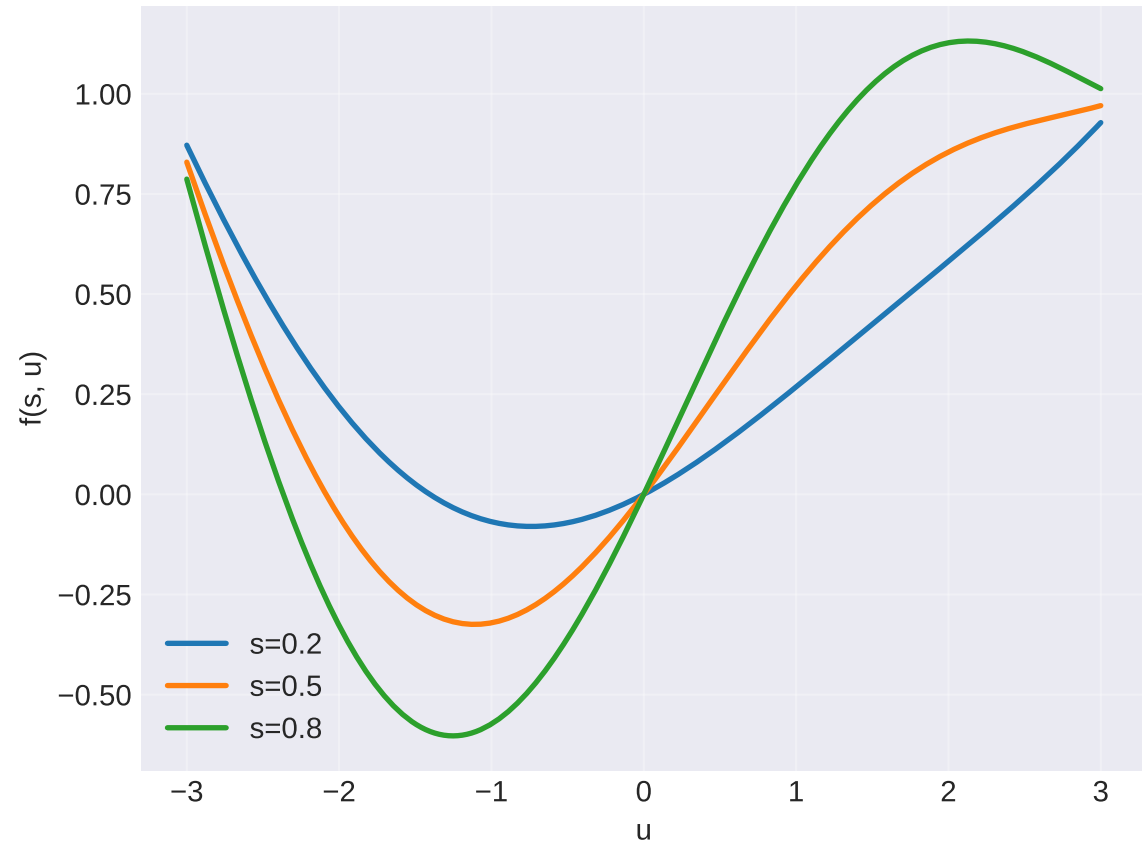
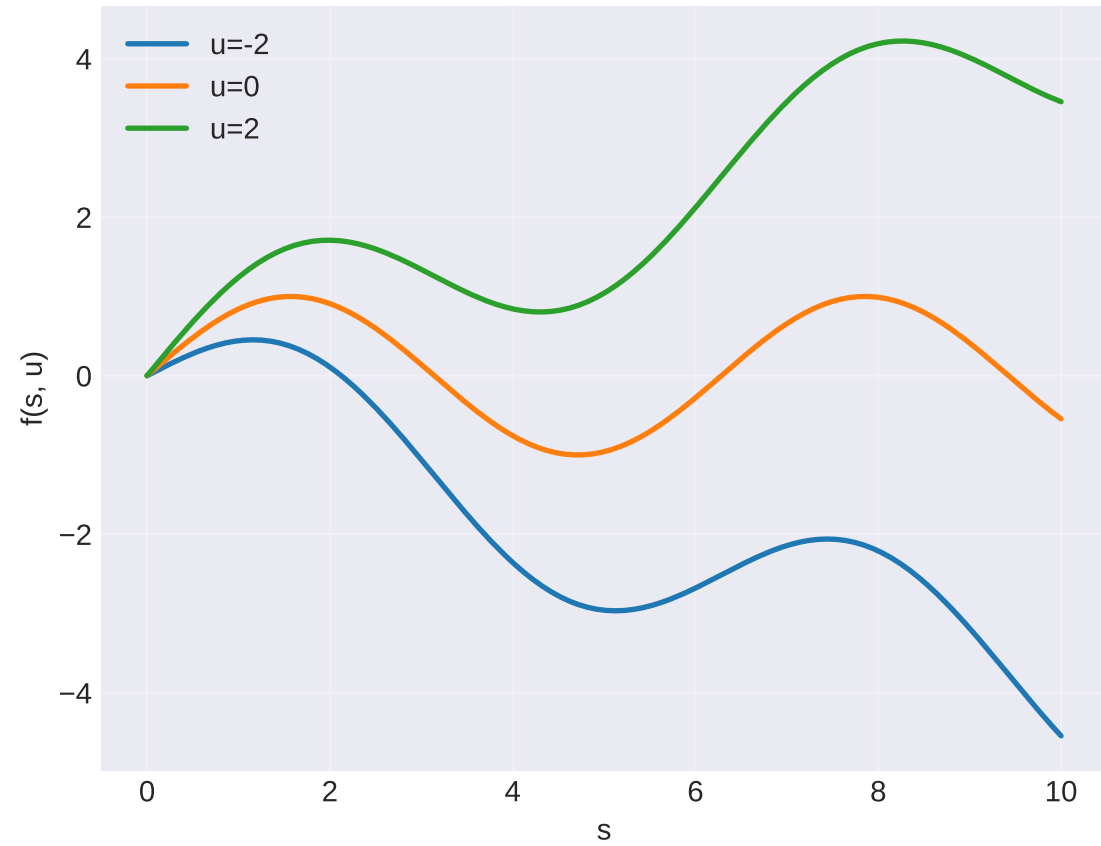


Carathéodory Conditions Illustrated

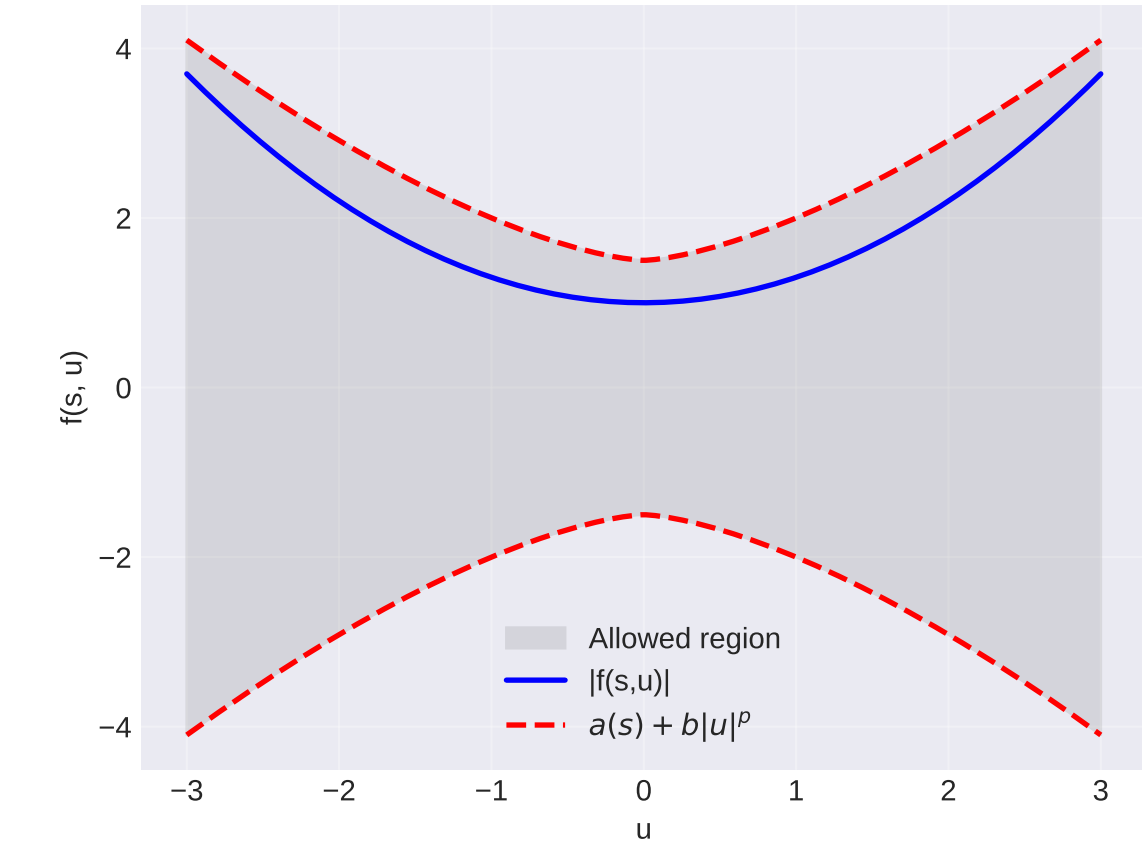
Condition 1: $f(s, \cdot)$ continuous for a.e. s



Condition 2: $f(\cdot, u)$ measurable for all u



Condition 3: Growth $|f(s, u)| \leq a(s) + b|u|^p$



Carathéodory Function: $f: \Omega \times \mathbb{R} \rightarrow \mathbb{R}$

Conditions:

(1) $f(s, \cdot)$ is continuous for almost all $s \in \Omega$

(2) $f(\cdot, u)$ is measurable for all $u \in \mathbb{R}$

(3) Growth: $|f(s, u)| \leq a(s) + b|u|^p$
where $a \in L_q(\Omega)$, $b \geq 0$
and $\frac{1}{p} + \frac{1}{q} = 1$

Result: Superposition operator F is continuous
from $L_p(\Omega)$ to $L_q(\Omega)$